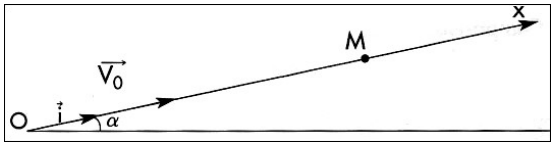
وزارة التربية والتعليم العالي المديرية العامة للتربية دائرة الامتحانات	امتحانات شهادة الثانوية العامة فرع علوم الحياة	دورة سنة 2003 الاستثنائية
	مسابقة في الفيزياء المدة : ساعتان	الاسم : الرقم :

This exam is formed of three obligatory exercises in three pages numbered from 1 to 3.
The use of non-programmable calculators is allowed.

<u>First Exercise</u> (6 points) Graphical study of energy exchange



Consider an inclined plane that makes an angle α with the horizontal ($\sin\alpha = 0.2$) and a marble (B) of mass m = 100 g, taken as a particle. We intend to study the energy exchange between the system (marble, Earth) and the surroundings.

To do that, the marble (B) is given, at the instant $t_o = 0$, the velocity $\vec{V_o} = V_o\vec{i}$ along the line of greatest slope OX. Given $V_o = 4\ m.s^{-1}$ and $g = 10\ m/s^2$.

The horizontal plane through point O is taken as the gravitational potential energy reference.

A- The forces of friction are supposed negligible.

1- Determine the value of the mechanical energy M.E of the system (marble, Earth).

2- At the instant t, the marble passes through a point M of abscissa OM = x. Determine, as a function of x, the expression of the gravitational potential energy $P.E_g$ of the system (marble, Earth) when the marble passes through M.

3-a) Trace, on the same system of axes, the curves representing the variations of the energies M.E and $P.E_g$ as a function of x.

Scale: - on the axis of abscissas: 1 cm represents 1 m;
- on the axis of energy: 1 cm represents 0.2 J.

b) Determine, using the graph, the speed of the marble for x = 3 m.

c) Determine, using the graph, the value of x_m of x for which the speed of (B) is zero.

B-1. In reality, the speed of the marble becomes zero at a point of abscissa x = 3 m. The forces of friction are no longer negligible. Calculate then the work done by the forces of friction between x = 0 and x = 3 m.

2. The system (marble, Earth) thus exchanges energy with its surroundings. In what form and by how much?

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This exam is formed of three obligatory exercises in three pages numbered from 1 to 3.

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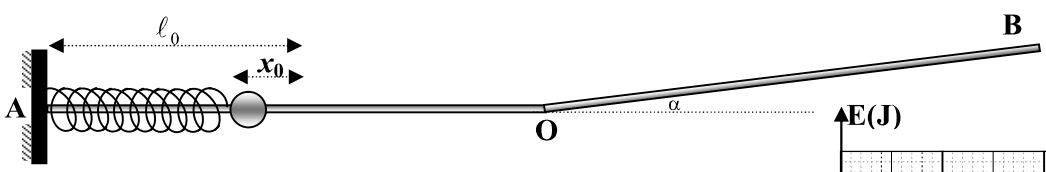
First Exercise (6 ½ pts) Determination of the value of a force of friction

A solid (S) of mass m = 200 g is free to move on a track AOB lying in a vertical plane. This rail is formed of two parts: the first one AO is straight and horizontal and the other OB is straight and inclined by an angle α with respect to the horizontal (sin α = 0.1). Along the part AO, (S) moves without friction, and along the part OB, (S) is acted upon by a force of friction $\vec{f}$ that is assumed constant and parallel to the path.

The object of this exercise is to determine the magnitude f of the force $\vec{f}$ of friction.

A- Launching the solid

In order to launch this solid on the part AO, we use a spring of constant k = 320 N / m and of free length ℓ_0 ; one end of the spring is fixed at A to a support. We compress the spring by x_o; we place the solid next to the free end of the spring and then we release them. When the spring attains its free length ℓ_0, the solid leaves the spring with the speed V_0 = 8 m/s; it thus slides along the horizontal part and then rises up at O the inclined part OB.



1) Determine the value of x_o.

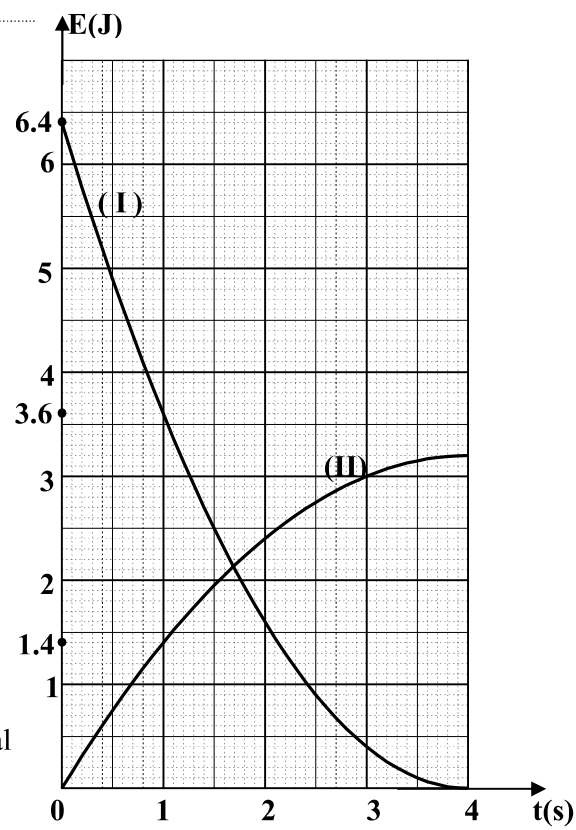
2) The solid reaches O with the speed V_0 = 8 m/s. Justify.

B- Motion of the solid along the inclined part OB

(S) moves, at O, up the inclined part OB with the speed V_0 at the instant t_o= 0. A convenient apparatus is used to trace, as a function of time, the curves representing the variations of the kinetic energy K.E of the solid and the gravitational potential energy $P.E_g$ of the system (solid - Earth). These curves are represented in the adjacent figure between the instants t_o= 0 and t_4 = 4 s, according to the scale:

- 1 division on the time axis corresponds to 1 s
- 1 division on the energy axis corresponds to 1 J.

The horizontal plane through point O is taken as a gravitational potential energy reference. Take g = 10 m / s^2.



1) The curve **(I)** represents the variation of the kinetic energy K.E of (S) as a function of time. Why?
2) Using the curves:

a) specify the form of the energy of the system at the instant $t_4 = 4$ s. Justify your answer.

b) determine the maximum distance covered by the solid along the part OB.

c)i) complete the table with the values of the mechanical energy M.E of the system at each instant t.

t (s)	0	1	2	3	4
M.E (J)		5			

ii) justify the existence of a force of friction $\vec{f}$.

iii) calculate the variation in the mechanical energy of the system between the instants $t_o = 0$ and $t_4 = 4$ s.

iv) determine f .

<u>Second Exercise</u> (7pts) <u>Determination of the inductance of a coil</u>

In order to determine the inductance L of a coil of negligible resistance, we put this coil with a resistor of resistance R = 10 Ω and a capacitor of capacitance $C = \frac{160}{\sqrt{3}}$ μF all in series across a generator G of adjustable frequency delivering an alternating sinusoidal voltage $u_{AM} = U_m \sin(2\pi f t)$ (figure 1). The circuit thus carries an alternating sinusoidal current i.

An oscilloscope is connected so as to display the voltage u_{AM} on the channel Y_1 and the voltage u_{DM} on the channel Y_2.

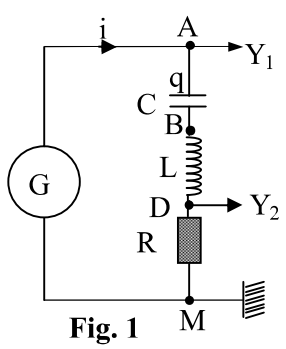


Fig. 1

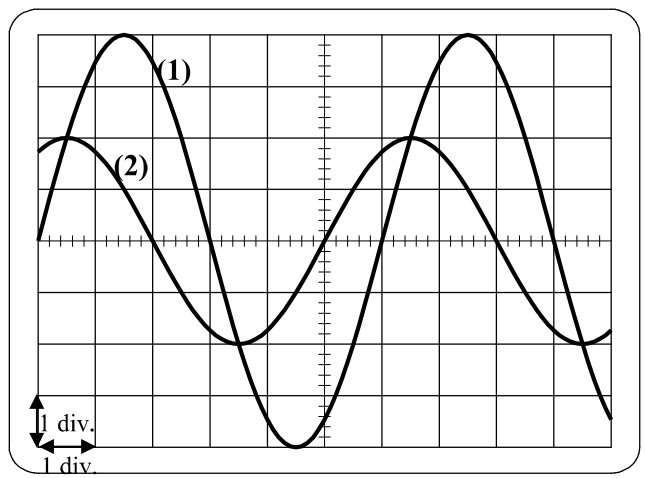


Fig. 2

A) The frequency of u_{AM} is adjusted at f = 50 Hz.

The waveforms of figure 2 show the curve (1) corresponding to the voltage u_{AM} and the curve (2) that corresponds to the voltage u_{DM}. The vertical sensitivity on both channels is 5 V / division.

Given: $\sqrt{3} = 1.73$ o.32 $\pi = 1$

1) Referring to the waveforms:

a) calculate the maximum voltage U_m across the generator.

b) show that the expression of the voltage u_{DM} may be written in the form:

$$u_{DM} = 10 \sin\left(100\pi t + \frac{\pi}{3}\right) \quad (u_{DM} \text{ in V, t in s}).$$

2) a) Determine the expression of i.

b) Show that the expression of the voltage across the terminals of the capacitor can be written as:

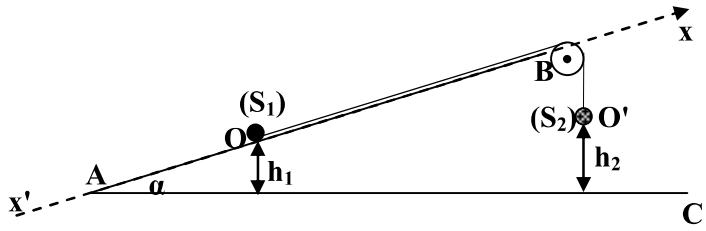
$$u_C = u_{AB} = -20\sqrt{3}\cos\left(100\pi t + \frac{\pi}{3}\right)$$

c) Determine the expression of the voltage u_{BD} across the coil in terms of the inductance L and time t.

Second Exercise *(7 points)*

Acceleration of a particle

The object of this exercise is to determine the expression of the magnitude of the acceleration of a particle using two methods. The apparatus used is formed of two particles (S_1) and (S_2) of respective masses m_1 and m_2, fixed at the extremities of an inextensible string passing over the groove of a pulley. (S_1), (S_2), the string and the pulley form a mechanical system (S).
The string and the pulley have negligible mass.
(S_1) may move on the line of greatest slope AB of an inclined plane that makes an angle α with the horizontal AC and (S_2) hangs vertically.
At rest, (S_1) is found at point O at a height h_1 above AC and (S_2) is found at O' at a height h_2 (adjacent figure).



At the instant $t_0 = 0$, we release the system (S) from rest. (S_1) ascends on AB and (S_2) descends vertically.
At an instant t, the position of (S_1) is defined by its abscissa $x = \overline{OS_1}$ on an axis x'Ox confounded with AB, directed from A to B.
Take the horizontal plane containing AC as a gravitational potential energy reference.
Neglect all the forces of friction.

1) Energetic method

a) Write down, at the instant $t_0 = 0$, the expression of the mechanical energy of the system [(S), Earth] in terms of m_1, m_2, h_1, h_2 and g.

b) At the instant t, the abscissa of (S_1) is x and the algebraic value of its velocity is v. Determine, at that instant t, the expression of the mechanical energy of the system [(S), Earth] in terms of m_1, m_2, h_1, h_2 , x, v, α and g.

c) Applying the principle of conservation of mechanical energy, show that :

$$v^2 = \frac{2(m_2 - m_1 \sin\alpha)gx}{(m_1 + m_2)}.$$

d) Deduce the expression of the value a of the acceleration of (S_1).

2) Dynamical method

a) Redraw a diagram of the figure and show, on it, the external forces acting on (S_1) and on (S_2). (The tension in the string acting on (S_1) is denoted by $\vec{T}_1$ of magnitude T_1 and that acting on (S_2) is denoted by $\vec{T}_2$ of magnitude T_2).

b) Applying the theorem of the center of mass $\Sigma\vec{F}_{ext} = m\vec{a}$,on each particle, determine the expressions of T_1 and T_2 in terms of m_1 , m_2 , g , α and a .

c) Knowing that $T_1 = T_2$, deduce the expression of a.